

SMART WINDOW AUTOMATION

Adithya M S (NSS22EC014), Alan Sunny (NSS22EC025), Anshif K (NSS22EC036), Krishnadas T R (NSS21EC044)

Guide - Asst Prof. Lija Arun, HOD - Prof. Dr. Sreeleja N Unnithan, Coordinator - Prof. Vinod G

Department Of Electronics and Communication Engineering NSS COLLEGE OF ENGINEERING Palakkad

ABSTRACT

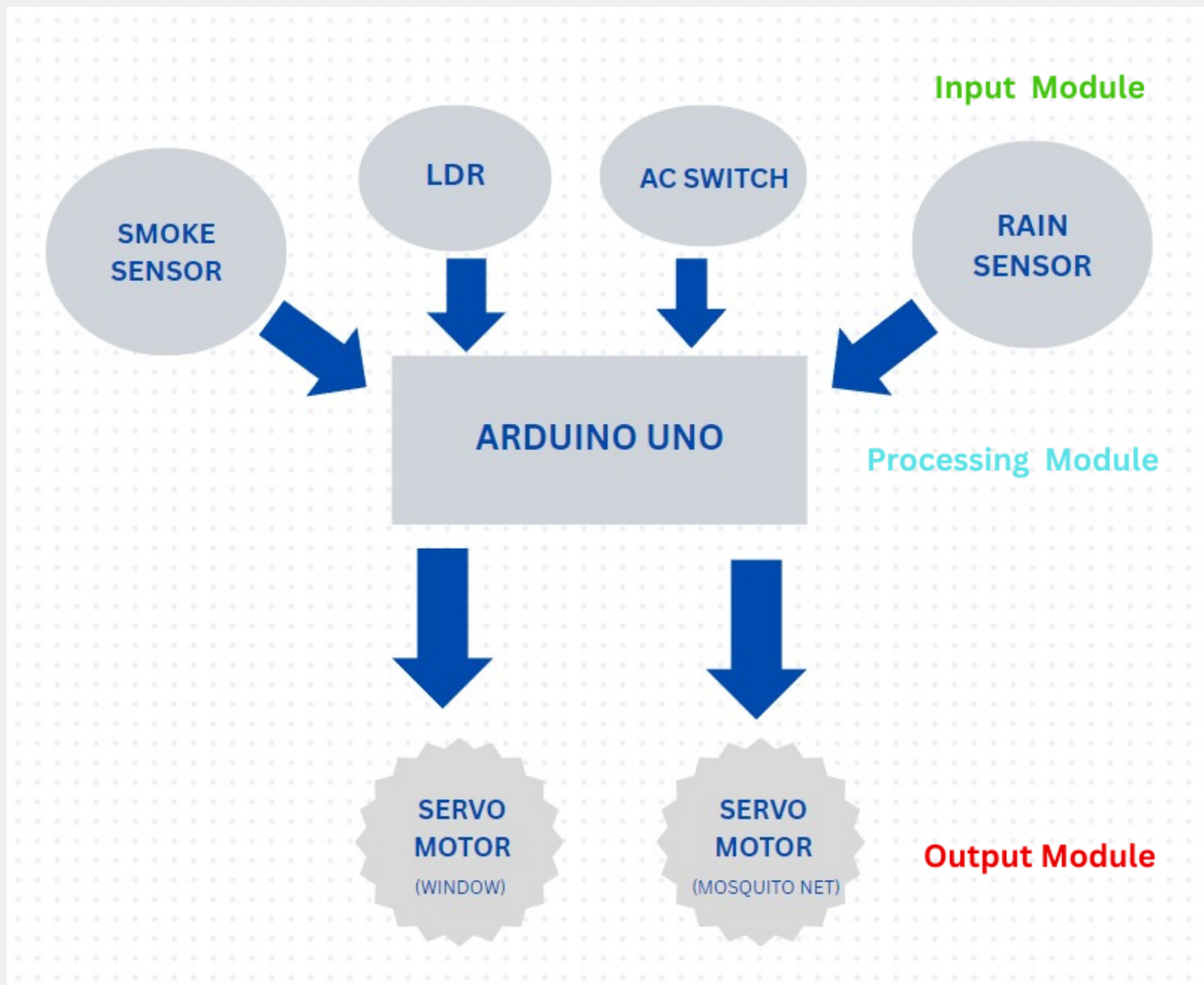
Our project focuses on enhancing home automation and environmental adaptation through an intelligent Smart Window System using Arduino Uno. This system autonomously controls the window based on environmental conditions such as rain, smoke, gas leakage, AC status, and light levels. When gas leakage or smoke is detected inside the house, the window automatically opens to ensure proper ventilation. In contrast, when it starts raining or the AC is turned on, the window closes to prevent water entry and maintain cooling efficiency. Additionally, the mosquito net deploys automatically in low light conditions to prevent insects from entering. By integrating multiple sensors, the system ensures a responsive and efficient operation, enhancing home safety, comfort, and energy efficiency. With cost-effective components and intelligent automation, this innovation creates a smarter and safer living environment.

INTRODUCTION

In today's world, smart home automation is becoming increasingly essential for enhancing safety, convenience, and energy efficiency. Our project, the Smart Window System, integrates Arduino Uno with multiple sensors to create an intelligent, automated window control mechanism. This system responds dynamically to environmental conditions, ensuring an adaptive and comfortable indoor atmosphere. By utilizing rain sensors, smoke detectors, gas sensors, LDRs, and AC status monitoring, the system effectively opens or closes the window based on real-time conditions. If smoke or gas leakage is detected inside the house, the window opens automatically to allow ventilation. Conversely, the window closes when it detects rain or when the AC is turned on, preserving indoor climate efficiency. Additionally, a mosquito net operates autonomously, remaining closed in low light to prevent insect entry and retracting in bright light. By integrating automation with smart decision-making, this project enhances home safety, energy efficiency, and environmental adaptability. With a cost-effective and userfriendly design, our system brings modern technology into everyday life, ensuring a safer and smarter living environment

CIRCUIT DIAGRAM

- **Input Module:** Handles data collection from sensors and user inputs.
- **Processing Unit:** Processes the data and applies the necessary algorithms.
- **Output Module:** Controls physical devices like window and mosquito net using servo motors.



COMPONENTS USED

Arduino UNO
Rain Sensor
Resistors
Smoke Sensor
LDR (Light Dependent Resistor)
Servo motors
Switch
Power Supply Unit

METHODOLOGY

Literature Survey: Review existing research and literature on automated window and mosquito net control systems.

Component Selection: Choose appropriate sensors and actuators, such as rain sensors, smoke sensors, LDR, servo motors, and switches, ensuring compatibility with Arduino Uno.

Circuit Design & Simulation: Design the circuit layout and simulate the system using software before hardware implementation.

Algorithm Development: Develop an algorithm to control the window and mosquito net based on sensor inputs and predefined conditions.

Validation and Testing: Test the system under various environmental conditions to ensure reliable operation.

Deployment: Integrate the system into a prototype and evaluate its performance.

Benefits of the Smart Window and Mosquito Net System

- **Automatic Window Operation:** Opens or closes based on environmental conditions like rain, smoke, and AC status.
- **Rain Protection:** Detects rain and closes the window automatically to prevent water from entering.
- **Smoke and Gas Alert System:** Opens the window when smoke or gas is detected to ventilate and ensure safety.
- **AC Optimization:** Closes the window when the AC is ON to maintain indoor cooling and save energy.
- **Light-Based Mosquito Net Control:** Deploys or retracts the mosquito net automatically based on ambient light levels.
- **Energy Efficiency:** Reduces energy waste by automating decisions that impact room temperature and ventilation.
- **Hands-Free Operation:** Fully automatic – no need for manual window or net adjustments.
- **Low Cost and Easily Scalable:** Built using affordable components, and can be implemented in homes, hostels, or offices.
- **User Safety and Comfort:** Improves indoor air quality and ensures protection from environmental elements.
- **Future Scope for IoT Integration:** Can be easily upgraded with IoT features like mobile app alerts and remote control.
- **Compact and Lightweight Design:** The system is compact and easy to install, making it ideal even for small rooms or rental spaces without permanent modifications.
- **Customizable for User Needs:** The system logic can be easily modified based on user preferences, such as sensitivity levels for rain or smoke, or setting preferred window positions.

DEVELOPED MODEL

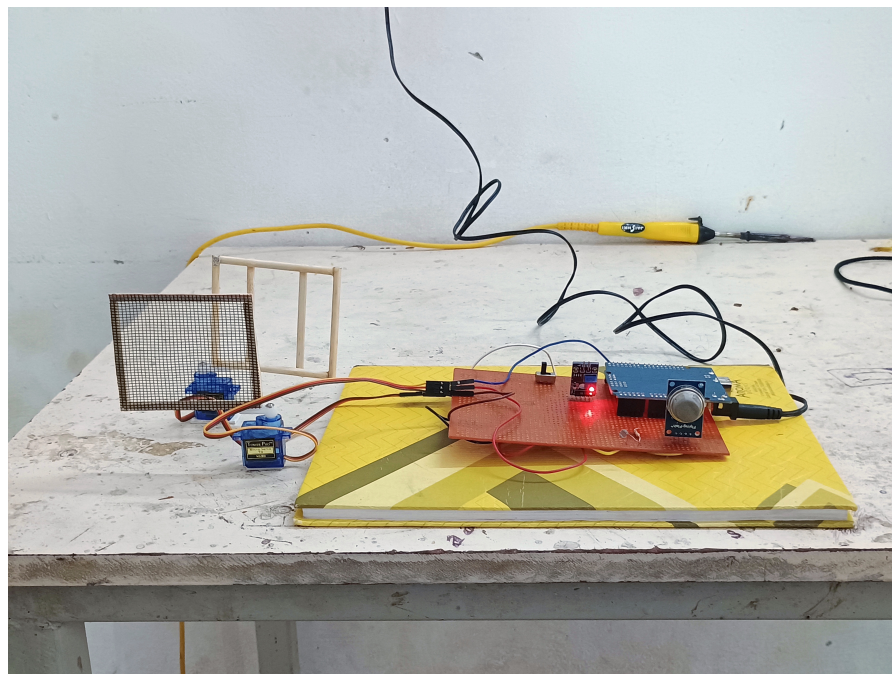


Figure 2: Developed prototype of the automated window and mosquito net control system

PROJECT SUMMARY

This project introduces an innovative Smart Window Automation System that integrates Arduino Uno with multiple environmental sensors to intelligently manage window operations and enhance indoor comfort and safety. Leveraging inputs from a rain sensor, smoke sensor, LDR (Light Dependent Resistor), and AC control switch, the system automatically opens or closes the window and deploys a mosquito net based on real-time conditions.

Designed using servo motors and simple yet effective logic, the system detects rain or AC usage and ensures the window is closed to prevent water entry and maintain energy efficiency. Conversely, it detects smoke or gas leakage and immediately opens the window to allow safe ventilation. The mosquito net is controlled based on light intensity, ensuring it remains deployed at night and retracted during the day.

This smart solution removes the need for human intervention, offering convenience, energy savings, and safety. Built with affordable and widely available components, the system is scalable, user-friendly, and ideal for homes, hostels, and office spaces. It provides a practical step toward intelligent building automation, combining simplicity with functionality to meet modern living standards.

By automating window control with sensor-based intelligence, this project delivers a reliable and eco-friendly alternative to manual operation, improving both indoor living conditions and energy management.

RESULT

Thus, the Smart Window and Mosquito Net Control System enables automated environmental adjustments based on real-time sensor inputs, ensuring enhanced safety, comfort, and energy efficiency. The system intelligently opens the window when gas leakage or smoke is detected inside, while closing it during rain or when the air conditioner is on. The mosquito net remains closed in dim lighting and opens only in bright daylight, providing optimal protection. This automation minimizes manual effort and enhances indoor air quality and security, demonstrating the effectiveness of sensor-based smart home solutions

References

- [1] Stolojescu-Crisan, Cristina, Calin Crisan, and Bogdan-Petru Butunoi. "An IoT based smart home automation system." *Sensors*, vol. 21, no. 11, p. 3784, 2021.
- [2] Vijayan, D. S., et al. "Automation systems in smart buildings: a review." *Journal of Ambient Intelligence and Humanized Computing*, pp. 1--13, 2020.
- [3] Kodali, Ravi Kishore, et al. "IoT based smart security and home automation system." *2016 International Conference on Computing, Communication and Automation (ICCCA)*. IEEE, 2016.